

BCEA :: CHEAT SHEET

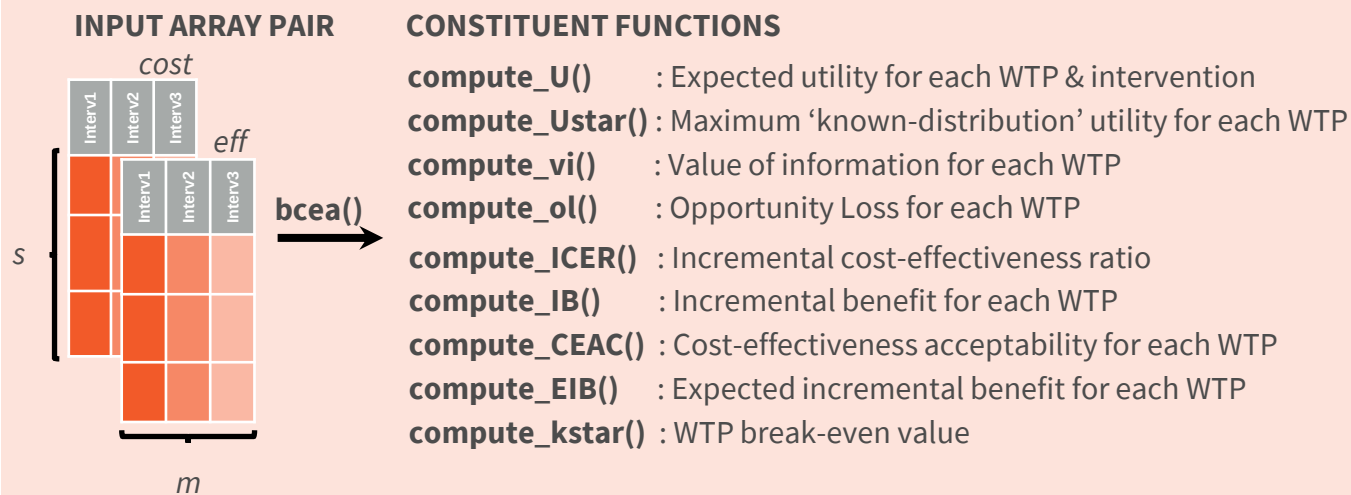


Introduction

Bayesian Cost-Effectiveness Analysis in R

Given a random sample of suitable variables of costs (*cost*) and clinical benefits (*eff*) for two or more interventions produces a health economic evaluation. Inputs may be the results of a Bayesian model (possibly based on MCMC) in the form of simulations from the posterior distributions. For *s* sample points compares one of the *m* interventions (*reference*) to the others (*.comparison*).

bcea(*eff*, *cost*, *ref*, *.comparison*, *interventions*)



bcea() calculates numerous cost-effectiveness analysis statistics. These can be called directly, using the constituent functions, but would require some pre-processing which is already handled by **bcea()**.

Value assignment

There are 3 equivalent ways to assign values to analysis parameters.

1. *In Constructor*: When first creating a **bcea** object.

```
he <- bcea(eff, cost, ref, .comparison, ...)
```

2. *Using Setters*: Change directly using replacement functions.

```
setComparison(he) <- comparison
setKmax(he) <- Kmax
setReference(he) <- ref
```

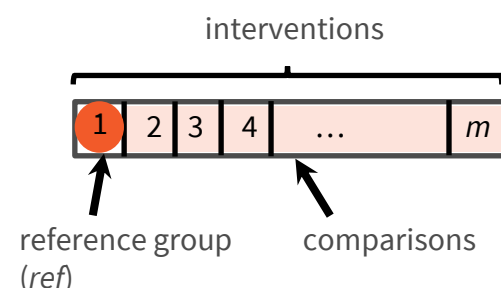
3. *In plotting call*: At the point of making a plot.

```
eib.plot(he, comparison, ...)
ceac.plot(he, comparison, ...)
ceplane.plot(he, comparison, ...)
```

SELECTING ANALYSIS INTERVENTIONS

Default

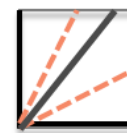
The first columns in (*eff*, *cost*) are the default reference intervention. All other interventions are the comparison interventions unless defined otherwise. E.g. for *m* interventions



Plot

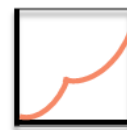
Standard cost-effectiveness analysis output plots. Base R, ggplot2 and plotly versions of plots are available and can be called directly but require extra default parameters.

Expected incremental benefit



eib.plot(*he*, *comparison* = NULL, *pos* = c(1, 0), *size* = NULL, *plot.cri* = NULL, *graph* = c("base", "ggplot2", "plotly"), ...) calls: • **eib_plot_base()** • **eib_plot_ggplot()** • **eib_plot_plotly()**

Expected value of information



evi.plot(*he*, *graph* = c("base", "ggplot2", "plotly"), ...)

Cost-effectiveness planes with contours



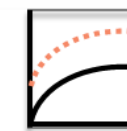
contour[2](*he*, *comparison* = 1, *scale* = 0.5, *nlevels* = 4, *levels* = NULL, *pos* = c(1, 0), *xlim* = NULL, *ylim* = NULL, *graph* = c("base", "ggplot2"), ...)

Compare optimal scenario to mixed case



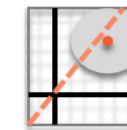
plot.mixedAn(*x*, *y.limits* = NULL, *pos* = c(0, 1), *graph* = c("base", "ggplot2"), ...)

Cost-effectiveness acceptability curve



ceac.plot(*he*, *comparison* = NULL, *pos* = c(1, 0), *graph* = c("base", "ggplot2", "plotly"), ...) calls: • **ceac_plot_base()** • **ceac_plot_ggplot()** • **ceac_plot_plotly()**

Cost-effectiveness plane



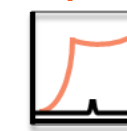
ceplane.plot(*he*, *comparison* = NULL, *pos* = c(1, 0), *graph* = c("base", "ggplot2", "plotly"), ...) calls: • **ceplane_plot_base()** • **ceplane_plot_ggplot()** • **ceplane_plot_plotly()**

Grid of CE plane, EIB, EVI & CEAC



plot.bcea(*x*, *comparison* = NULL, *pos* = c(1, 0), *graph* = c("base", "ggplot2", "plotly"), ...)

Expected value of perfect partial information



plot.evppi(*x*, *pos* = c(0, 0.8), *graph* = c("base", "ggplot2"), *col* = NULL, ...)

Summarise data

Summary statistics and formatted tables can be used to interrogate a **bcea()** object.

summary.bcea(*he*, ...)

Prints a table with summary results of the health economic evaluation

summary.mixedAn(*he*, ...)

Prints summary table for results of mixed analysis

sim.table(*he*, ...)

Summary table of simulations from the cost-effectiveness analysis

make.report(*he*, ...)

Constructs the automated report from the output of the BCEA